physicists Hans Reissner in 1916 and Gunnar Nordstrom in 1918. However, nobody understood the physical meaning of Reissner's and Nordström's solution until 1960, when two of Wheeler's students, John Graves and Dieter Brill, discovered that it describes a charged black hole.

Box 7.2

The Organization of Soviet and Western Science: Contrasts and Consequences

As I and my young physicist colleagues struggled to develop the hoop conjecture and to prove that black holes have no hair and to discover how they lose their hair, we also were discovering how very differently physics was organized in the U.S.S.R. than in Britain and America, and what profound effects those differences have. The lessons we learned may have some value in planning for the future, especially in the former Soviet Union, where all state institutions—scientific as well as governmental and economic—are now (1993) struggling to reorganize along Western lines. The Western model is not completely perfect, and the Soviet system was not uniformly bad!

In America and Britain there is a constant flow of young talent through a research group such as Wheeler's or Sciama's. Undergraduates may join the group for their last, senior year, but they then are sent away for graduate study. Graduate students join it for three to five years, and then are sent elsewhere for postdoctoral study. Postdocs join it for two or three years and then are sent away and expected either to start a research group of their own elsewhere (as I did at Caltech) or to join a small, struggling group elsewhere. Almost nobody in Britain or America, no matter how talented, is allowed to stay on in the nest of his or her mentor.

In the U.S.S.R., by contrast, outstanding young physicists (such as Novi-kov) usually remained in the nest of their mentor for ten, twenty, and sometimes even thirty or forty years. A great Soviet mentor like Zel'-dovich or Landau usually worked in an Institute of the Academy of Sciences, rather than in a university, so his teaching load was small or nonexistent; by keeping his best former students, he built around himself a